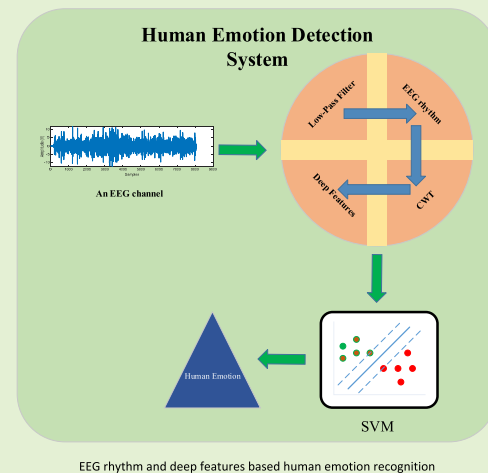


Exploring Deep Learning Features for Automatic Classification of Human Emotion Using EEG Rhythms

Fatih Demir¹, Nebras Sobahi², Member, IEEE, Siuly Siuly³, and Abdulkadir Sengur⁴

Abstract—Emotion recognition (ER) from Electroencephalogram (EEG) signals is a challenging task due to the non-linearity and non-stationarity nature of EEG signals. Existing feature extraction methods cannot extract the deep concealed characteristics of EEG signals from different layers for efficient classification scheme and also hard to select appropriate and effective feature extraction methods for different types of EEG data. Hence this study intends to develop an efficient deep feature extraction based method to automatically classify emotion status of people. In order to discover reliable deep features, five deep convolutional neural networks (CNN) models are considered: AlexNet, VGG16, ResNet50, SqueezeNet and MobilNetv2. Pre-processing, Wavelet Transform (WT), and Continuous Wavelet Transform (CWT) are employed to convert the EEG signals into EEG rhythm images then five well-known pretrained CNN models are employed for feature extraction. Finally, the proposed method puts the obtained features as input to the support vector machine (SVM) method for classifying them into binary emotion classes: valence and arousal classes. The DEAP dataset was used in experimental works. The experimental results demonstrate that the AlexNet features with Alpha rhythm produces better accuracy scores (91.07% in channel Oz) than the other deep features for the valence discrimination, and the MobilNetv2 features yields the highest accuracy score (98.93% in Delta rhythm (with channel C3) for arousal discrimination.

Index Terms—EEG based emotion classification, EEG rhythms, CWT, deep features, pretrained CNN models.



I. INTRODUCTION

EMOTION itself is constituted of neuro-physiological variations associated with thoughts, behavioral responses and a degree of pleasure or displeasure [1], [2]. Emotions, which is understood from facial expressions, have important role in

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human-to-human communication. Sounds and body gestures are the other important emotion indicators. The major part of the emotions in human's daily life is the management of attention [3], [4] and decision-making [5], [6]. Emotion detection can improve various artificial intelligence based, applications namely, patient monitoring, criminal detection disabled assistance, security services, robotics, and communication [7], [8]. There have been several methods for classification of emotions such as bio signals (e.g. EEG, electrocardiogram (ECG) and electromyography (EMG) etc.), imaging techniques and videos [8], [9]. The EEG data is defined as non-stationary time series, which refers to the recording of the brain's spontaneous electrical activity [10], [11]. Many research studies have been undertaken to investigate the effects of emotion on EEG signals and ER from EEG signals. Issa *et al.* [12] used CWT and broad learning system for EEG based emotion recognition (ER). After CWT, the gray scale image feature extraction was employed and three classifiers were employed for classification into four emotions. Atkinson and Campos [13] used fourteen EEG electrodes for a novel feature-based approach for EEG

based ER. Authors used various statistical parameters such as standard deviation, median and kurtosis. Beside statistical features, Hjorth parameters, band power and the fractal dimension were used for feature extraction. A feature selection mechanism was employed after feature extraction and the SVM classifier was preferred in classification stage of the proposed approach. Kumar *et al.* [14] used only 2 EEG electrodes for achieving the EEG based ER. The higher order spectrum analysis was used to obtain the bispectrum features on the EEG signals. Extracted features were classified by using the SVM classifier. Gupta *et al.* [15] used 6 electrodes and flexible analytic wavelet transform (FAWT) to perform EEG based ER. Authors used various information based features and SVM and random forest (RF) classifiers in their proposed work. Zhang *et al.* [16] used ontology for EEG based ER. The proposed method was depended on two steps. The users' contexts, EEG and the environmental parameters were considered in the first step and in the second one, modeling of the reasons on users' emotions were extracted. Zhang *et al.* [17] proposed an methodology for EEG based ER. The empirical mode decomposition (EMD) and sample entropy were used for feature extraction and SVM classifier was used for classification. Candra *et al.* [18] used the WT for EEG based ER. The Shannon entropy and SVM classifier was used for construction of the proposed method. Rozgić *et al.* [19] used an approach for EEG based ER which was composed of three steps. An overlapping window was used for feature extraction in the first step. In the second step, feature transformation was carried out by using the non-parametric nearest-neighbor model. Finally, the obtained features were classified. Al-Nafjan *et al.* [20] proposed a methodology for EEG based ER. Power spectral density and frontal asymmetry features and deep neural networks were considered in the proposed approach. Chen *et al.* [21] proposed convolutional neural networks (CNN) for EEG based ER. A CNN model was proposed and trained on the EEG signals. Zhuang *et al.* [22] proposed an approach where EMD was used EEG based ER. Energy and phase information were used in classification where SVM classifier was considered. Li *et al.* [23] used ensemble of SVM classifiers for EEG based ER. A scheme, which was composed of weighted fusion, was considered.

In this paper, a hybrid approach is proposed. The proposed approach employs EEG rhythms for ER. The EEG signals were initially low-pass filtered for denoising. Then, the WT is employed for rhythm extraction. The extracted EEG rhythm signals are then converted to the EEG rhythm images by using the CWT method. A series of pre-trained CNN models consisting of AlexNet, VGG16, ResNet50, SqueezeNet and MobilNetv2 are used to extract deep features [18]–[22]. The obtained deep features are used in SVM classifier for the classification process. The well-known DEAP dataset was considered for performance evaluation of proposed method [16].

The contributions of the proposed method are summarized as follows:

(1) An effective automated ER framework is presented which combines EEG rhythm, deep CNN features

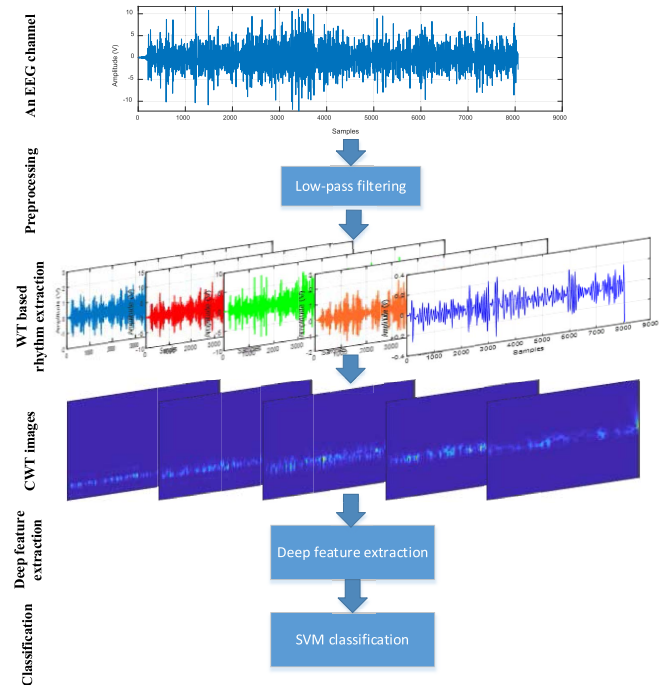


Fig. 1. Illustration of the proposed method.

and SVM to define valence and arousal emotions from EEG signals;

(2) Investigating the effectiveness of deep features through five deep learning CNN models: AlexNet, VGG16, ResNet50, SqueezeNet and MobilNetv2;

(3) Discovering appropriate deep features with appropriate EEG band and channel through statistical analysis for the SVM classifier;

The reminder of this paper is as following. Next section introduces the proposed methodology. The proposed method and the related theories were briefly introduced in the proposed methods section. Section 3 gives the dataset, experimental works and the results. The discussions on experimental works were given in Section 3. The conclusions were given in Section 4.

II. PROPOSED METHODOLOGY

Our proposed method is designed in an emotional classification model for recognizing valence and arousal. The valence is considered in 2 classes, low valence (LV) and high valence (HV) and arousal is considered as low arousal (LA) and high arousal (HA). Fig. 1 illustrates the framework of the proposed methodology. As seen in Fig. 1, the EEG signals are initially denoised by using a low-pass filtering. And, WT is employed for decomposition of the EEG signals into its rhythms. These rhythm signals are converted into rhythm images by employing the CWT. The constructed rhythm images are conveyed to the pretrained CNN models namely; AlexNet, VGG16, ResNet50, SqueezeNet and MobilNetv2 for deep feature extraction. Finally, the classification of the rhythm images into emotion categories is carried out by using the SVM classifier.

A. Pre-Processing: Noise Removing by Low-Pass Filtering

In general, the EEG data is very noisy and often affected by artifacts that can bias the analysis of the data and lead to incorrect conclusions. That's why, before going the analysis the EEG data, we first removed the noises and artifacts by applying of the low-pass filtering. The reason for employing low-pass filtering is that the low-pass filter helps to keep each EEG channel in its horizontal zone, eliminating large up or down shifts in the space of other channels. Using low-pass filtering we denoised the raw EEG signals and then used for the following step.

It is worth to mentioning that each EEG channel was investigated independently in order to determine the efficient EEG channel and EEG rhythm on ER. The pseudo-code of the proposed study is given in Table I.

B. EEG Rhythms Extraction

As EEG signals convey different emotional information in different frequency band, thus we divided EEG signals into five bands in the frequency scale in this study. We employed the WT to extract the rhythm of the full-band EEG signal. We performed a four level decomposition by using the Daubechies 4th order wavelet (db4) function [24]. In the first level decomposition, the detail coefficients indicate the Gamma rhythm (30-60 Hz). In the second level decomposition, the detail coefficients indicate the Beta rhythm (15-30 Hz). In the third level decomposition, the detail coefficients indicate the Alpha rhythm (8-15 Hz). In the last decomposition level, while the detail coefficients show the Theta rhythm (4-7 Hz), the approximation coefficients show the Delta rhythm (0-4 Hz). We extracted below five types of EEG rhythms by the WT from original EEG signals:

- Delta rhythm covers the frequency band less than 4 Hz (associated with Deep Dreamless Sleep, Loss of Body Awareness, quietness, lethargy, fatigue)
- Theta rhythm covers the frequency band greater than or equal to 4 and less than 8 Hz (Associated with Dreams, Deep Meditation, REM sleep, Creativity, awake state, or the emotion gradually becomes calmer),
- Alpha rhythm covers the frequency band greater than or equal to 8 and less than or equal to 14 (associated with Relaxation (while awake), Pre-sleep Drowsiness),
- Beta rhythm covers the frequency band greater than 14 and less than 40 Hz (associated with Active Thinking, Concentration, and Cognition)
- Gamma rhythm covers the frequency band greater than or equal to 40 Hz, (associated with Higher Mental Activity, Consciousness, Perception, multisensory information integration).

C. Converting EEG Rhythms to EEG Images

The CWT, which is a TF method, is employed for transforming the EEG rhythms into EEG images. And, it is defined as [25];

$$F_{\omega}(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t) \bar{\psi}\left(\frac{t-a}{b}\right) dt \quad (1)$$

TABLE I

THE PSEUDO-CODE OF THE PROPOSED EEG BASED EMOTION DETECTION METHOD

Input: Raw EEG signals
Output: ER
For all channels
For subjects
Denoising by using low-pass filtering
Wavelet Decomposition based rhythm extraction
Convert rhythm signals into rhythm images by using CWT
For rhythms
Extract deep features by using pretrained CNN models
end
Apply SVM classification
end
end

where, $\psi(t)$ is called as the mother wavelet. The mother wavelet is a wavelet function that is represented in both time and frequency domains. The over line symbolizes the complex conjugate operation. The Morse wavelet function is used in CWT.

D. Extraction of Deep Features

This study considered five pre-trained CNN models: AlexNet, VGG16, ResNet50, SqueezeNet and MobilNetv2 that were trained on 25 million of images. An investigation was performed about these five models, which one is better to produce higher performance in the ER. Table II shows some important properties of these CNN models. AlexNet is known as the first deep CNN model, which has gained much attention in artificial intelligence community. Its depth is 8 and the image input size is 227×227 [26]. The 'fc6' layer was used for feature extraction and the obtained deep feature vectors are 4096 dimensional. VGG16 was designed as a deeper CNN model compared to the AlexNet [27].

The depth of the VGG16 is 16 and the input size is 224×224 . The 'fc6' layer of the VGG16 was used for feature extraction and the obtained deep feature vectors are 4096 dimensional. The ResNet50 is a deeper model compared to AlexNet and VGG16 [28]. It was composed of 50 layers. Its image input size is 224×224 . The 'fc1000' layer was used for feature extraction and the extracted feature vectors are 1000 dimensional. SqueezeNet was developed as a fast and effective CNN model [29]. As its depth is 18 but it achieves AlexNet-level accuracy with 50x fewer parameters. The image input size of SqueezeNet is 227×227 . The 'pool10' layer is used for 1000 dimensional feature vector extraction. MobileNetv2, of which depth is 53, produces 1000 dimensional feature vector by using the 'logits' layer [30]. The input size of MobileNetv2 model is 224×224 .

E. Classification

The SVM classifier, which is a well-known machine learning approach, is considered in this study. The SVM classifier was preferred due to its high performance in wide range of its applications [31].

Let's assume a two class data denoted by $(x_1, y_1), (x_2, y_2), \dots, (x_i, y_i)$ and $i = 1, 2, \dots, M$. x_i

TABLE II
SOME PROPERTIES OF THE PRETRAINED CNN MODELS

Pretrained CNN model	Depth	Image input size	Feature length
AlexNet	8	227×227	4096
VGG16	16	224×224	4096
ResNet50	50	224×224	1000
SqueezeNet	18	227×227	1000
MobileNetv2	53	224×224	1000

shows the input feature vector and y shows the class labels. Thus, Eq. (2) can be used for obtaining of the hyper-plane that used to separate the two classes.

$$f(x) = w^T x + b = 0 \quad (2)$$

where, w and b indicates the weight vector and the bias value used to determine the position of the hyper-plane, respectively. SVM was initially developed for the datasets that can be separated linearly. If the dataset cannot be separated linearly, a kernel function is employed to turn dataset into another hyper-plane. To this end, different kernel functions such as polynomial and radial basis function can be used. SVM employs an optimization procedure to detect the optimum separating hyper-plane, which is given in Eq. (3).

$$\begin{cases} \min \frac{\|w\|^2}{2} \\ y_i (w^T x_i + b) \geq 1 \quad i = 1, 2, \dots, M \end{cases} \quad (3)$$

Eq. (3) can be revised by adding C regularization parameter and a positive artificial ξ_i variable as follows:

$$\min f(w, \xi) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^M \xi_i \quad (4)$$

III. EXPERIMENTAL WORKS

A. DEAP Dataset

The DEAP data set, which contains 32-electrode EEG signals from 32 subjects, was constituted while each subject was watching forty one-minute music videos. The subjects rated each music video in terms of valence, arousal, like / dislike, dominance, and familiarity levels. In addition to the EEG signals, a facial video was recorded for 22 of the 32 subjects while watching each music video. Ratings of subjects during the experiment were carefully analyzed. Correlations among EEG signal frequencies and subjects' evaluations were also analyzed. EEG signals were recorded with a sampling frequency of 512 Hz and then down-sampled up with a sampling frequency of 128 Hz. Thus, a size of $40 \times 32 \times 8064$ data was constituted for each subject.

B. Results and Discussions

For experiments, a computer, which has Intel (R) CPU (3.20 GHz) with 32 GB of memory, was utilized. The extracted features were divided into training and test sets in the ratio of 0.75 and 0.25, respectively. This process was carried out in a random fashion. The SVM classifier was preferred due to its high performance in wide range of applications [31].

TABLE III
CLASSIFICATION ACCURACY FOR ALEXNET MODEL IN HV VS LV FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
Oz	8.21%	84.29%	91.07%	80.36%	82.86%	85.34%
F3	86.79%	85.00%	79.64%	80.71%	80.36%	82.50%
P4	83.93%	80.71%	83.21%	77.14%	82.14%	81.43%
PO3	89.29%	77.14%	80.36%	82.86%	79.29%	81.79%
CP5	80.36%	76.07%	89.64%	78.21%	82.50%	81.36%
Average	85.72%	80.64%	84.78%	79.86%	81.43%	

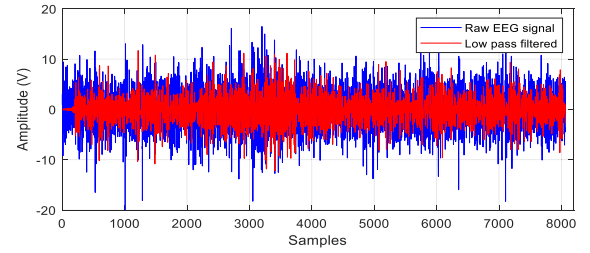


Fig. 2. The raw and low-pass filtered of an EEG signal.

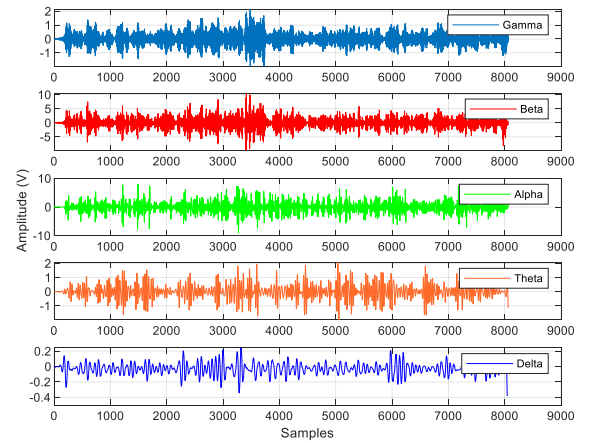


Fig. 3. The rhythms of an EEG signal.

Also, it determines the maximum-margin hyperplane to reduce the prediction error. The hyperparameter C was searched in the range of $[10^{-4}-10^3]$. The radial basis function kernel was used in SVM and its parameters were tuned by using the grid search optimization algorithm. Double tagging was considered during the labeling of emotions. Valence ratings below five were assumed to contain negative emotions, and valence scores higher than five were considered to have positive feelings. In addition, arousal rating scales range from passive to active. Arousal ratings less than five were considered passive, and other rating scales higher than five were considered active. Therefore, the class labels obtained were high arousal (HA) with low valence (LV) and high valence (HV) and low arousal (LA), respectively.

Fig. 2 illustrates a raw signal and low-pass filtered signal of an EEG signal and Fig. 3 shows the gamma, beta, alpha, theta and delta rhythms of an EEG signal, respectively.

Fig. 4 shows the EEG rhythm images that were obtained by using the CWT. The x-axis shows the time and the y-axis shows the frequency. As seen in Fig. 4, each rhythm is covering a frequency band.

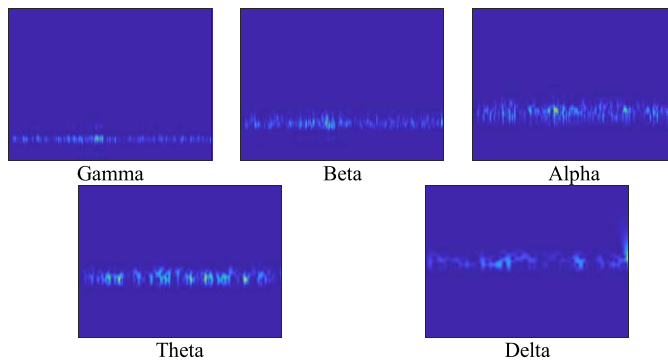


Fig. 4. The rhythm images of an EEG signal that are obtained via CWT.

TABLE IV

CLASSIFICATION ACCURACY FOR VGG16 MODEL IN HV VS LV FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
FC5	76.07%	82.86%	84.29%	82.50%	82.50%	81.64%
PO3	87.50%	80.00%	81.43%	76.79%	80.71%	81.29%
Cz	82.50%	81.07%	83.21%	74.64%	85.00%	81.28%
F3	84.64%	76.07%	87.86%	76.07%	80.71%	81.07%
C3	85.71%	82.14%	75.00%	82.86%	79.64%	81.07%
Average	83.28%	80.43%	82.36%	78.57%	81.71%	

TABLE V

CLASSIFICATION ACCURACY FOR RESNET50 MODEL IN HV VS LV FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
Cz	78.21%	86.43%	80.36%	73.57%	82.86%	80.29%
F3	84.29%	78.93%	83.57%	73.93%	80.36%	80.22%
Fp2	77.50%	84.29%	78.57%	79.64%	77.14%	79.43%
FC5	80.36%	73.57%	77.50%	81.43%	83.93%	79.36%
FC1	84.64%	72.86%	81.07%	72.50%	85.00%	79.21%
Average	81.00%	79.22%	80.21%	76.21%	81.86%	

The initial experiments were conducted on HV vs LV classification and the obtained all results were illustrated in Fig. 4. In Fig. 4 each cell shows the achievement of a pretrained CNN model.

The x-axis, y-axis and z-axis of the given plots show the rhythms, EEG channels and accuracy scores, respectively. For the x-axis, alpha, beta, delta, gamma and theta rhythms are denoted by the numbers 1, 2, 3, 4 and 5, respectively. Moreover, Tables III-VII show the achievements of the five EEG channels on all EEG rhythms. These channels were selected due to the highest average accuracy scores of the rhythms achievements. In Table III, the performances of AlexNet features are provided based on different channels and bands. As observed from Table III, Oz channel and Delta rhythm produced a 91.07% accuracy score, which was the highest accuracy score that AlexNet features have produced.

The second highest accuracy score 89.64% was produced by the CP5 and Delta rhythm and the third highest accuracy score 89.29% was obtained by the PO3 and Alpha rhythm, respectively. From Table III, it can be inferred that Alpha rhythm and Oz channel were produced better achievements with AlexNet and SVM classifier.

TABLE VI

CLASSIFICATION ACCURACY FOR SQUEEZE NET MODEL IN HV VS LV FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
AF4	78.93%	84.29%	76.43%	82.50%	81.79%	80.79%
Oz	89.64%	73.57%	80.00%	72.14%	83.21%	79.71%
AF3	89.29%	83.57%	71.43%	75.71%	77.50%	79.50%
T8	81.07%	82.14%	80.00%	82.14%	71.07%	79.28%
P4	80.00%	76.79%	83.21%	81.07%	74.64%	79.14%
Average	83.79%	80.07%	78.21%	78.71%	77.64%	

TABLE VII

CLASSIFICATION ACCURACY FOR MOBILNETV2 MODEL IN HV VS LV FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
Cz	86.43%	83.57%	76.43%	84.64%	83.93%	83.00%
Oz	86.79%	83.93%	83.93%	78.93%	80.36%	82.79%
F3	91.07%	82.50%	79.29%	78.57%	82.50%	82.79%
P3	78.57%	81.79%	86.43%	85.00%	78.57%	82.07%
PO3	85.00%	81.43%	84.64%	73.57%	82.14%	81.36%
Average	85.57%	82.64%	82.14%	80.14%	81.50%	

TABLE VIII

CLASSIFICATION ACCURACY FOR ALEXNET MODEL IN HV VS LV FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
Fp2	86.43%	93.93%	93.57%	93.93%	83.21%	90.21%
T7	84.64%	90.00%	92.86%	96.43%	85.36%	89.86%
P7	83.57%	86.79%	91.79%	95.00%	89.29%	89.29%
FC5	86.43%	90.00%	90.36%	92.14%	85.36%	88.86%
F7	83.57%	88.57%	88.21%	93.93%	87.50%	88.36%
Average	84.93%	89.86%	91.36%	94.29%	86.14%	

Table IV gives the achievements of the VGG16 features and SVM classifier on HV vs LV discrimination. As seen in Table IV, the Alpha rhythm and the FC5 channel produced the highest average accuracy scores. With VGG16 features, the best achievement 87.86% accuracy score was produced by F3 channel and Delta rhythm.

In Table V, the accuracy scores, which were obtained by ResNet50 features, were given for each rhythm and EEG channel, respectively. According to the average accuracy scores, Theta rhythm and Cz channel produced the highest average accuracy scores. While Cz produced an average 80.29% accuracy score, Theta rhythm obtained an average 81.86% accuracy score. The best achievement 86.43% was produced by the Cz channel and Beta rhythm.

In Table VI, the achievements of EEG channels against rhythms with SqueezeNet features on HV vs LV classification were given. As seen in Table VI, the best average rhythm achievement 83.79% was obtained by the Alpha rhythm and the highest average accuracy score 80.79% was produced by the AF4 channel, respectively. The best accuracy score 89.64% was obtained by the Alpha rhythm and Oz channel. The mobilNetv2 features achievements were represented in Table VII. As observed in Table VII, alpha rhythm and Cz channel produced a 85.57% and a 83.00% average accuracy scores respectively. The best achievement with MobilNetv2 features was a 91.07%, and it was obtained by F3 channel and Alpha rhythm.

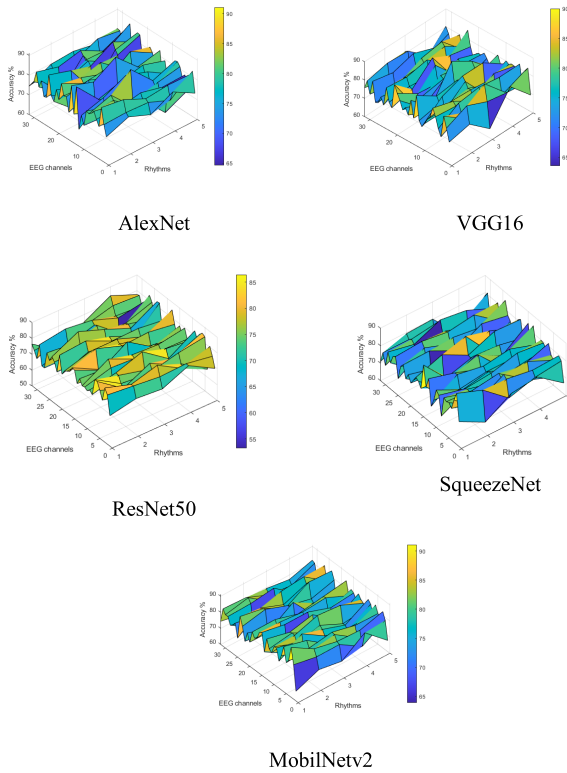


Fig. 5. The performance of deep feature + SVM on LV and HV classification.

Experiments were also conducted on HA vs LA classification and the obtained all results with deep features were illustrated in Fig. 5 similar to the Fig. 4. Tables VIII–XII show the achievements of the five EEG channels that produced the highest average accuracy scores of the rhythms achievements. Table VIII shows the achievements of the AlexNet features and SVM. As observed from Table VIII, Gamma rhythm and Fp2 channel produced a 94.29% and 90.21% accuracy scores, respectively, that were the highest average accuracy scores. The best accuracy score 96.43% was also produced by the Gamma rhythm and T7 channel.

Table IX represents the obtained accuracy scores for VGG16 features. From Table IX, it is seen that Beta and Theta rhythms obtained average accuracy scores above 90.00% and the CP5 channel also produced a 90.29% average accuracy score, respectively. A 94.64% accuracy score, which the highest among all results, was produced by the CP5 channel and Theta rhythm.

Table X shows the achievements of the ResNet50 features and SVM on HA vs LA classification. As seen in Table X, Delta rhythm and P7 channel produced a 92.36% and 91.50% accuracy scores, respectively, that were the highest average accuracy scores. The best accuracy score 93.93% was also produced by the Delta rhythm and P7 channel and Gamma rhythm and FC1 channel, respectively.

Table XI shows the performance of the SqueezeNet features on HA vs LA classification. As seen in Table XI, Theta rhythm and P7 channel produced a 90.86% and 90.64% accuracy

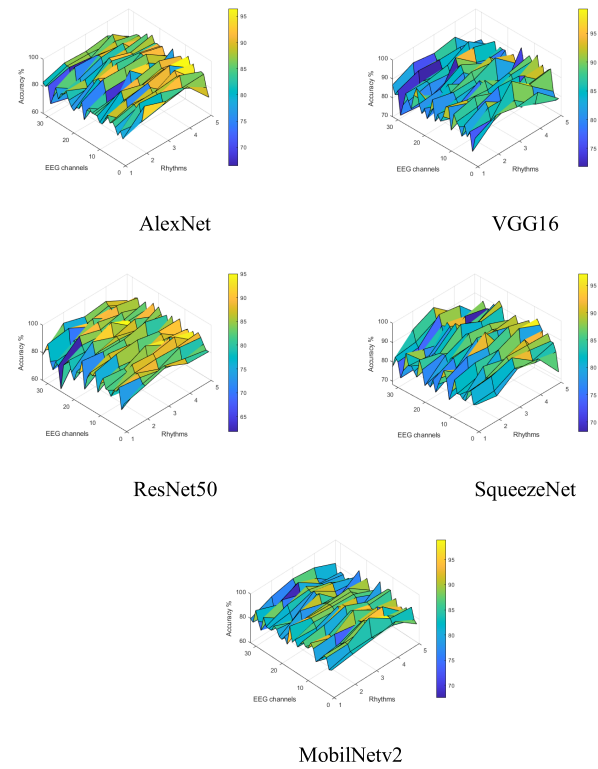


Fig. 6. The performance of deep feature + SVM on LA and HA classification.

TABLE IX

CLASSIFICATION ACCURACY FOR VGG16 MODEL IN HA vs LA FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
CP5	90.36%	92.86%	88.21%	85.36%	94.64%	90.29%
P7	91.79%	88.57%	87.14%	89.29%	91.79%	89.72%
F7	82.50%	91.07%	90.71%	90.00%	91.43%	89.14%
FC5	87.86%	93.57%	86.43%	88.57%	90.00%	89.29%
Fp2	91.43%	88.57%	88.57%	93.21%	83.57%	89.07%
Average	88.79%	90.93%	88.21%	89.29%	90.29%	

scores, respectively, that were the highest average accuracy scores. The best accuracy score 96.43% was also produced by Gamma rhythm and T7 channel, respectively.

Table XII shows the performance of the MobilNetv2 features on HA vs LA classification. As seen in Table XII, Delta rhythm and P7 channel produced a 94.93% and 90.57% accuracy scores, respectively, that were the highest average accuracy scores. The best accuracy score 98.93% was also produced by Delta rhythm and F3 channel, respectively.

In Table XIII, the proposed method was compared with the existing methods that use the same dataset. As shown in Table XIII, the proposed method outperformed the existing other methods on HV vs LV categorization. Alzarzi *et al.* produced a second-best accuracy score where the accuracy was 85.8% [32]. Tripathi *et al.* [33] and Li *et al.* [23] reported 81.4% and 80.7% accuracy scores, respectively. Rozgić *et al.* [19], Zhang *et al.* [17], and Atkinson and Campos [13] reported accuracy scores among 76.9%, 75.2%

TABLE X

CLASSIFICATION ACCURACY FOR RESNET50 MODEL IN HA VS LA FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
P7	89.29%	92.86%	93.93%	90.71%	90.71%	91.50%
Fp2	91.43%	88.93%	88.57%	91.79%	87.50%	89.64%
FC6	86.07%	84.64%	95.00%	82.86%	90.36%	87.79%
CP5	88.21%	86.07%	93.21%	84.64%	86.79%	87.78%
FC1	78.57%	92.50%	91.07%	93.93%	81.79%	87.57%
Average	86.71%	89.00%	92.36%	88.79%	87.43%	

TABLE XI

CLASSIFICATION ACCURACY FOR SQUEEZE NET MODEL IN HA VS LA FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
P7	86.43%	91.43%	90.00%	94.29%	91.07%	90.64%
FC6	88.57%	85.71%	94.29%	89.29%	90.71%	89.71%
T7	81.79%	84.29%	91.79%	96.43%	91.79%	89.22%
CP5	88.21%	90.71%	90.00%	82.14%	94.29%	89.07%
F3	87.14%	88.21%	91.43%	83.57%	86.43%	87.36%
Average	86.43%	88.07%	91.50%	89.14%	90.86%	

TABLE XII

CLASSIFICATION ACCURACY FOR MOBILNETV2 MODEL IN HA VS LA FOR DIFFERENT RHYTHMS WITH RESPECT TO EACH EEG CHANNELS

EEG channel	Alpha	Beta	Delta	Gamma	Theta	Average
P7	89.29%	91.43%	96.07%	88.21%	87.86%	90.57%
F3	88.57%	88.57%	98.93%	87.50%	85.71%	89.86%
CP5	92.14%	85.36%	96.79%	85.71%	84.29%	88.86%
Fp2	94.64%	90.36%	88.57%	87.50%	80.00%	88.21%
FC5	85.00%	86.43%	94.29%	82.86%	82.50%	86.22%
Average	89.93%	88.43%	94.93%	86.36%	84.07%	

TABLE XIII

PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH SOME OF THE STATE-OF-ART-RESULTS

Method	Accuracy (%)	
	HV vs LV	HA vs LA
Koelstra <i>et al.</i> [35]	57.6	62.0
Alazrai <i>et al.</i> [32]	85.8	86.6
Huang <i>et al.</i> [34]	66.1	82.5
Candra <i>et al.</i> [18]	65.1	65.3
Rozgic <i>et al.</i> [19]	76.9	69.1
Zhang <i>et al.</i> [17]	75.2	81.7
Atkinson <i>et al.</i> [13]	73.1	73.0
Tripathi <i>et al.</i> [33]	81.4	73.3
Zhuang <i>et al.</i> [22]	69.1	71.9
Li <i>et al.</i> [23]	80.7	83.7
Proposed study	91.1	98.9

and 73.1% and Zhuang *et al.* [22], Huang *et al.* [34], and Candra *et al.* [18] reported performance of their method among 60.0% and 70.0%. The worst result reported by Koelstra *et al.* [35] was 57.6%.

The achievement comparisons on HA and LA categorization were also shown in Table XIII. As shown in Table XIII, the proposed method obtained the best classification score. The second-best result was produced by Alazrai *et al.* [32]. Huang *et al.* [34], Zhang *et al.* [17], and Li *et al.* [23] reported accuracy results between 80.0% and 85.0%, respectively. Atkinson and Campos [13], Tripathi *et al.* [33], and

TABLE XIV

2D EMOTION CLASSIFICATION LABELS

Emotion Label	Scores
HVHA	$V \geq 5, A \geq 5$
LVHA	$V < 5, A \geq 5$
LVLA	$V < 5, A < 5$
HVLA	$V \geq 5, A < 5$

Zhuang *et al.* [22] reported accuracy results between 70.0% and 75.0% respectively. Also, Koelstra *et al.* reported the worst accuracy score [35].

In order to further evaluate the performance of the proposed method, we performed a multiclass classification between four different classes of emotions: HVHA, LVHA, LVLA, and HVLA in the same DEAP database. The labelling process is given in Table XIV. These results were obtained for Oz channel, AlexNet features and Alpha rhythm. The SVM was also tuned as mentioned previously.

The best accuracy was 90.6% for HVHA emotion state, 89.23% for LVHA emotion state, 83.1% for LVLA emotion state, and 83.4% for HVLA emotion state.

IV. CONCLUSION

This study intended to develop an emotion classification framework that can use EEG signal data for recognizing valence and arousal from humans. The proposed framework consists of several steps such as noise removing by low-pass filtering; EEG rhythms extraction by the WT, the rhythms conversation to EEG images by the CWT, deep feature extraction and investigation by the five CNN models and the SVM classification of the extracted deep features. The followings are important conclusions of the proposed method.

1) In the HV vs LV discrimination, the AlexNet features with Alpha rhythm produced better average accuracy scores than the other deep features. Besides, the Oz channel with AlexNet features produced the best average classification score. The MobilNetv2 features achieved the second best performance for HV vs LV classification. It is surprising that an initial model (AlexNet) produced better results than the recent deep models.

2) In the HV vs LV classification, generally Cz, F3, PO3 and Oz channels produced considerable results than the other channels.

3) In the HA vs LA discrimination, the AlexNet features with Gamma rhythm produced better average accuracy scores than the other deep features. Besides, the P7 channel with ResNet50 features produced the best average classification score. The MobilNetv2 features achieved the second best performance for HV vs LV classification. The highest accuracy score 98.93% was obtained by the F3 channel, Delta rhythm and MobilNetv2 features.

4) The proposed method produced better results in HA vs LA classification than HV vs LV classification.

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